IE 692: Optimization Methods for Reinforcement Learning

Course Project Checkpoint

Proposal: *Fri, Nov 10th (11.59pm EST)* Final due date: *Tue, Nov 11th (11:59pm EST)*

1. Team and Logistics

Name	Email	Attendance
Jack Morris	jmorr185@vols.utk.edu	In-person
Anjila Budathoki	abudatho@vols.utk.edu	In-person
Manish Dhakal	mdhakal1@vols.utk.edu	In-person

All team members confirm that they are available to present on Tuesday, November 24th from 12:55pm to 2:10pm EST.

2. Proposed Enhancements

We propose three enhancements: (i) SPMD with three Bregman divergences, (ii) True Online TD(λ) Evaluation, and (iii) entropy-regularized SPMD.

2.1 SPMD with Three Bregman Divergences

Category: Policy optimization

Algorithm 1 Stochastic Policy Mirror Descent

procedure OPTIMIZATION(K, π_0)
  $\pi_k \leftarrow \pi_0$
  for $k \leftarrow 0$ **to** K **do**
    $\widetilde{Q}^{\pi_k} \leftarrow \widetilde{Q}^{\pi_k}(s,a) \approx Q^{\pi}(s,a)\ \forall\, s \in \mathcal{S},\ a \in \mathcal{A}$
    $\pi_k \leftarrow \underset{\pi_k(\cdot|s) \in \Delta_{|\mathcal{A}|}}{\arg\min} \left\{ \left\langle \widetilde{Q}^{\pi_k}(s,\cdot),\, \pi(\cdot \mid s) \right\rangle \eta_k + \boldsymbol{D}(\pi_k(\cdot \mid s) \,\|\, \pi(\cdot \mid s)) \right\}$
  end for
  return π_k
end procedure

Description. We instantiate SPMD with three Bregman divergences generated by different distance-generating functions: squared Euclidean distance, KL divergence, and Tsallis divergence. For example, when KL divergence is used as $\boldsymbol{D}(\pi_k(\cdot \mid s) \,\|\, \pi(\cdot \mid s))$ (In Algorithm 1) we get

$$\underset{p \in \Delta_{|\mathcal{A}|}}{\arg\min} \left\{ \left\langle \widetilde{Q}^{\pi_k}(s,\cdot),\, p \right\rangle \eta_k + \sum_{a \in \mathcal{A}} p_a \ln \frac{p_a}{\pi_k(s,a)} \right\}$$

as our update to π_k.

Motivation. We aim to measure the convergence and performance of the learned π_k under each divergence for each MDP problem. Where there is an observed difference in either metric, we will identify empirically or derive its source.

TODOS.

- We are yet to work in bregman divergences.

2.2 True Online TD(λ): Tabular vs. Linear Approximation

Category: Policy evaluation

Description. Based on the feedback on proposal and lectures on policy evaluation methods, we compare conventional TD(λ) and True Online TD(λ) for evaluating the action-value function of each fixed stochastic policy π_k generated by Stochastic Policy Mirror Descent (SPMD). These methods are implemented in tabular and linear function-approximation settings, giving four TD configurations. We compare all these four against Truncated On-Policy Monte Carlo (TOMC) evaluator as the project baseline.

Let $\phi(s,a) \in \mathbb{R}^d$ denote a state–action feature vector, and let $\Phi \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|\times d}$ be the feature matrix. The action-value estimate is

$$\widetilde{Q}_\theta(s,a) = \phi(s,a)^\top \theta, \qquad \theta \in \mathbb{R}^d. \tag{1}$$

For a fixed policy π_k, the projected Bellman operator is given:

$$F(\theta) = \Phi^\top W\big(\Phi\theta - c - \gamma P^{\pi_k}\Phi\theta\big), \qquad \theta_{t+1} = \theta_t - \eta F(\theta_t), \tag{2}$$

where c is the state–action cost vector, P^{π_k} is the state–action transition matrix induced by π_k, and W is a diagonal weighting matrix. In the continuing, stationary setting, W may be formed using $w(s,a) = \nu^{\pi_k}(s)\pi_k(a \mid s)$.

In the online setting, instead of forming the full operator, we use the observed transition $(s_t, a_t, c_t, s_{t+1}, a_{t+1})$, where $a_{t+1} \sim \pi_k(\cdot \mid s_{t+1})$. Write $\phi_t = \phi(s_t, a_t)$ and $\phi_{t+1} = \phi(s_{t+1}, a_{t+1})$. The single-transition stochastic operator is

$$\widetilde{F}_t(\theta_t) = \phi_t\left(\phi_t^\top\theta_t - c_t - \gamma\phi_{t+1}^\top\theta_t\right), \qquad \theta_{t+1} = \theta_t - \eta\widetilde{F}_t(\theta_t). \tag{3}$$

Motivation. The required TOMC baseline uses sampled truncated returns to estimate Q^{π_k}. In contrast, TD methods update after each observed transition by bootstrapping from the next estimate.

Hypothesis.

- We expect TD(λ) to reduce the policy evaluation error faster than TOMC because it updates the value estimates after each transition and incorporates information from previous updates using exponentially decaying weights.
- For tabular and linear function approximation, we expect linear approximation to learn faster initially because it shares information across different state-action pairs. However, tabular learning may achieve better final accuracy because it estimates each state-action value separately, without introducing feature approximation error.

Evaluation Metrics. Plot independently validated policy cost $V^{\pi_k}(s_0)$ against cumulative environment observations. In addition, when a reference action-value vector Q^{π_k} is available for validation, plot the weighted prediction error

$$E_Q(t) = \|\Phi\theta_t - Q^{\pi_k}\|_W \tag{4}$$

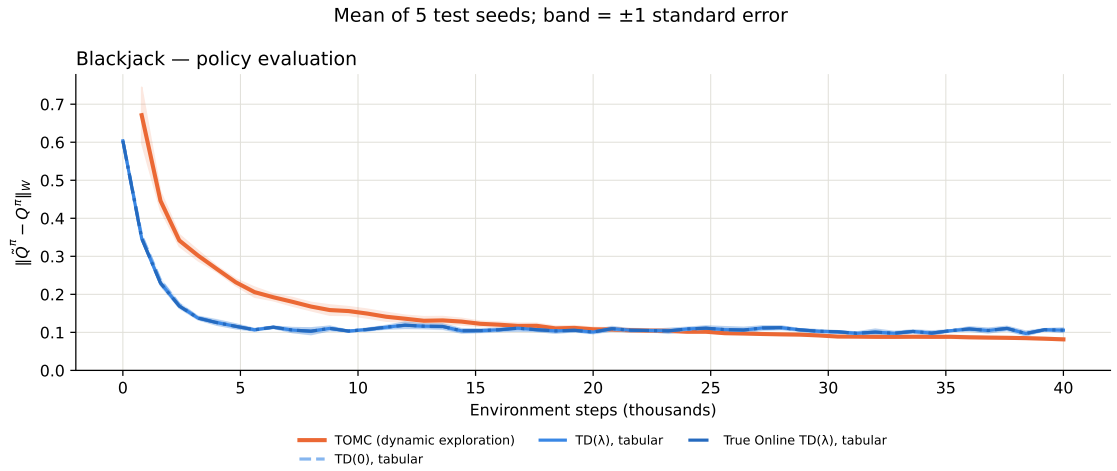


Figure 1: This plot is for Blackjack. We observe that TD learns faster early but levels off near 0.10, TOMC overtakes it at about 20k steps and ends lower (0.082 vs. 0.106). The three TD curves overlap because no state–action pair repeats within a hand.

Results.

Table 1: Blackjack, fixed uniform policy: weighted action-value error (mean ± standard error over 5 test seeds). Lower is better; the lowest error at each step count is in bold.

Steps	TOMC	TD(0)	TD(λ)	True Online TD(λ)
800	0.671 ± 0.073	0.354 ± 0.004	$\mathbf{0.347 \pm 0.004}$	$\mathbf{0.347 \pm 0.004}$
2,400	0.342 ± 0.016	0.173 ± 0.005	$\mathbf{0.169 \pm 0.005}$	$\mathbf{0.169 \pm 0.005}$
8,000	0.168 ± 0.011	$\mathbf{0.103 \pm 0.009}$	$\mathbf{0.103 \pm 0.009}$	$\mathbf{0.103 \pm 0.009}$
20,000	0.108 ± 0.005	$\mathbf{0.100 \pm 0.004}$	0.101 ± 0.004	0.101 ± 0.004
40,000	$\mathbf{0.082 \pm 0.003}$	0.105 ± 0.006	0.106 ± 0.006	0.106 ± 0.006

As shown in Table 1, all TD methods achieve lower policy evaluation errors than TOMC during the early stages of learning in Blackjack. After 800 observations, the TD methods achieve an error of 0.347, compared to 0.671 for TOMC, with lower variation across random seeds. This is likely because TD methods update value estimates after each transition, whereas TOMC must wait until a complete coverage round before updating its estimates. However, TD stops improving at an error of about 0.10 after roughly 8,000 steps, and TOMC overtakes it at about 20,000 steps, finishing with the lowest error (0.082 vs 0.106). We attribute this to TD's constant step size, which leaves it at a fixed noise level, whereas TOMC's averaging over completed rounds keeps reducing its error. TD(0), TD(λ) and True Online TD(λ) perform almost identically, because Blackjack hands are short and no state-action pair repeats within a hand, so eligibility traces and the true-online correction have little effect. We therefore expect these methods to show clearer differences in GridWorld, where trajectories are longer and repeated visits are common.

TODOS.

- Run all methods, tabular and linear, on GridWorld, and run the linear methods on Blackjack.
- Following the Nov 10 demonstration, add the linear TD fixed point as a reference line to separate approximation error from sampling error.

- Decide whether to compare against GAE, or explain how it relates to TD(λ).
- Tune α, λ, and the TOMC horizon T with a 5–10% budget and separate tuning seeds.
- Run SPMD with each evaluator and plot $V^{\pi_t}(s_0)$ and the number of observations per iteration.
- Use the same baseline curve as the Entropy-Regularized SPMD section, and fix the TD and TOMC formulas in the write-up.

Reference.

2.3 Entropy-Regularized SPMD

Category: Policy optimization

Description. We add a negative-entropy regularizer $\tau\, h(p)$, $h(p) = \sum_a p_a \log p_a$, to the SPMD subproblem (2.2.21),

$$\pi_{k+1}(\cdot \mid s) = \operatorname*{arg\,min}_{p \in \Delta_{|\mathcal{A}|}} \ \eta_k \big[\langle \widetilde{Q}(s,\cdot),\, p\rangle + \tau\, h(p)\big] + \mathrm{KL}(p \,\|\, \pi_k(\cdot \mid s))\,,$$

which is the entropy-regularized problem of Section 2.1.2. With the KL divergence the update stays closed-form,

$$\pi_{k+1}(a \mid s) \ \propto \ \pi_k(a \mid s)^{\frac{1}{1+\eta_k \tau}} \exp\left(-\frac{\eta_k\, \widetilde{Q}(s,a)}{1+\eta_k \tau}\right),$$

and $\tau = 0$ recovers baseline SPMD (2.2.3). The regularizer pulls the policy toward the Boltzmann policy $\pi \propto \exp(-Q^\pi/\tau)$, so every action keeps a probability bounded away from zero.

Motivation. Online policy evaluation needs every action to keep positive probability (2.3.5), but unregularized SPMD converges toward a deterministic policy, so rarely chosen actions stop being sampled and their value estimates go stale. On GridWorld this can lock the policy into a route that never reaches a goal. Entropy regularization slows this collapse, at the price of converging to the regularized optimum.

Hypothesis. $\tau > 0 \to$ higher $\min_a \pi_k(a \mid s) \to$ more distinct (s,a) pairs sampled per observation $\to$ lower error in $\widehat{Q}$ at rarely-visited states $\to$ better improvement direction $\to$ faster decrease than unregularized $V^{\pi_k}(s_0)$ per observation.

Results.

- **GridWorld:** $\tau = 0$ and $\tau = 0.001$ drop fastest and stay below the baseline until about 1.5M steps. After that the baseline catches up and ends level with $\tau = 0.001$. Larger τ settles early at a fixed, worse level: about 0.05 gap for $\tau = 0.01, 0.08$ for $\tau = 0.03$ and 0.11 for $\tau = 0.1$.
- **Blackjack:** the baseline, $\tau = 0$ and $\tau = 0.001$ all reach the optimal policy within about 5k steps. $\tau = 0.001$ has two brief jumps around 55k steps. $\tau = 0.1$ never settles and swings between gaps of about 0.02 and 0.07.

Observations.

- With limited budget of iterations/updates, we observe that entropy regularization reduces the cost rapidly, closing the cost gap *w.r.t* the optimal policy.
- Overscaling the entropy regularization introduces noisy output of the policy. The optimization methods are highly sensitive to τ, the choice of τ has to be made with extensive care; usually lower τ gives better values.

TODOs.

- Test an annealed schedule $\tau_k = \tau_0/(k+1)^p$ that keeps early exploration without the bias.
- Run ablation analysis and complementary effects with other Bregman divergences.

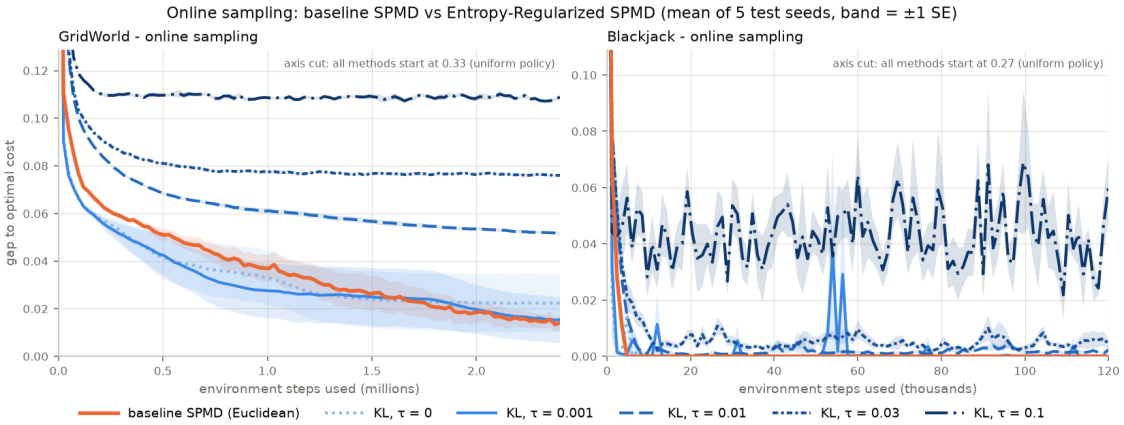


Figure 2: We observe rapid decrease in the gap to the optimal cost in the earlier iteration but they eventually converge to the same value with larger iterations. Overscaling the entropy regularization, i.e., $\tau \in \{0.01, 0.03, 0.2\}$, leads to noisy policy, especially for Blackjack problem.